Affinity Drifting: Learning Conditional Distributions from One Sample per Condition

Chan Lee

Working draft — results update automatically as experiments finish (2026-11-09 03:35 UTC, commit `ea07857`)

Abstract

Drifting models train a one-step generator by moving its samples along a field $\mathbf{V}_{p,q} = \mathbf{V}^+_p - \mathbf{V}^-_q$ that attracts them toward positive samples from the data distribution p and repels them from negative samples of the generated distribution q. In text-to-image data, however, each caption usually comes with a single image, so the empirical conditional $p_{\text{data}}(\cdot \mid c)$ is a point mass. We show that the attraction toward a point mass reduces exactly to ℓ_2 regression, so a generator that cannot resolve nearby conditions collapses to the conditional mean. We propose Affinity Drifting, which draws positives from a Nadaraya–Watson estimate of the conditional distribution, pooling the samples of semantically similar conditions with weights from a fixed condition embedding, while keeping the sample-space kernel of drifting. For Gaussian kernels the field is a scaled difference of smoothed log-densities, which identifies its fixed points: with negatives from the condition's own generated samples the equilibrium is the Nadaraya–Watson mixture, while pooling the negatives in the same way memorizes the training positives. We test this with pre-registered, one-factor-at-a-time experiments. In a non-parametric particle system the predicted fixed points are reached at the resolution of the kernel. In a 2-D toy with known conditional distributions, the original Drifting objective and the one-positive objective both collapse to the conditional mean (ratio of total variances $\operatorname{tr}\boldsymbol{\Sigma}_q / \operatorname{tr}\boldsymbol{\Sigma}_p$ of 2.8×10^{-5} and 3.8×10^{-4}), while an oracle with many samples per condition does not (0.909). Affinity Drifting restores the mixture: it captures every mode and lowers the held-out Wasserstein-2 distance to 0.33 of the one-positive objective, on par with the oracle. Ablations show why: pooling the positives without the sample-space kernel still collapses, and random instead of semantically similar neighbors place samples off the conditional support. Contrary to our prediction, the gap does not close when the condition is given to the generator at full resolution: the one-positive generator regresses to the conditional mean even on its training conditions.

1. Introduction

Drifting models [2] train a one-step generator f_θ by evolving its pushforward distribution $q_\theta = (f_\theta)_{\#}p_{\boldsymbol{\epsilon}}$ during training rather than at sampling time: each generated sample is attracted toward positive samples $\mathbf{y}^+ \sim p$ and repelled from negative samples $\mathbf{y}^- \sim q_\theta$, and the generator regresses onto its own drifted outputs. With class labels every class has many images, and the attraction is a kernel mean-shift toward the class-conditional distribution $p_{\text{data}}(\cdot \mid c)$.

Text conditioning breaks this assumption. A caption typically describes one image, so the empirical $p_{\text{data}}(\cdot \mid c)$ is a single point mass. The normalized kernel weight of that one positive is then identically one, the attraction no longer depends on where the sample is, and drifting reduces to ℓ_2 regression onto the positive (Lemma 1). A generator of finite resolution in c averages the positives of the conditions it cannot distinguish and outputs their mean, a blurred sample between modes (Fig. 1).

We ask whether pooling the positives of semantically similar conditions restores a distribution, and why. Affinity Drifting replaces the point mass by a Nadaraya–Watson estimate of $p_{\text{data}}(\cdot \mid c)$ and keeps the sample-space kernel $k(\mathbf{x}, \mathbf{y}^+)$ of drifting, so the attraction becomes a mean-shift toward a mixture rather than a pull toward an average. The claim is mechanistic: the gain should come from the $\mathbf{x}$-dependent assignment of samples to modes, not merely from having more targets. We test it with ablations that remove one alternative explanation at a time: pooling positives with a constant sample-space kernel, pooling random rather than similar neighbors, and perturbing the single positive with matched noise, against an oracle with true conditional samples and the original Drifting objective as a baseline.

Contributions. (i) The one-positive degeneracy of drifting and its consequence, mean collapse, which in our toy occurs even when the generator sees the condition at full resolution. (ii) Affinity Drifting, with a characterization of its fixed points for Gaussian kernels. (iii) A pre-registered, one-factor-at-a-time evaluation, run from a non-parametric particle system to a neural 2-D toy. This draft reports fin-

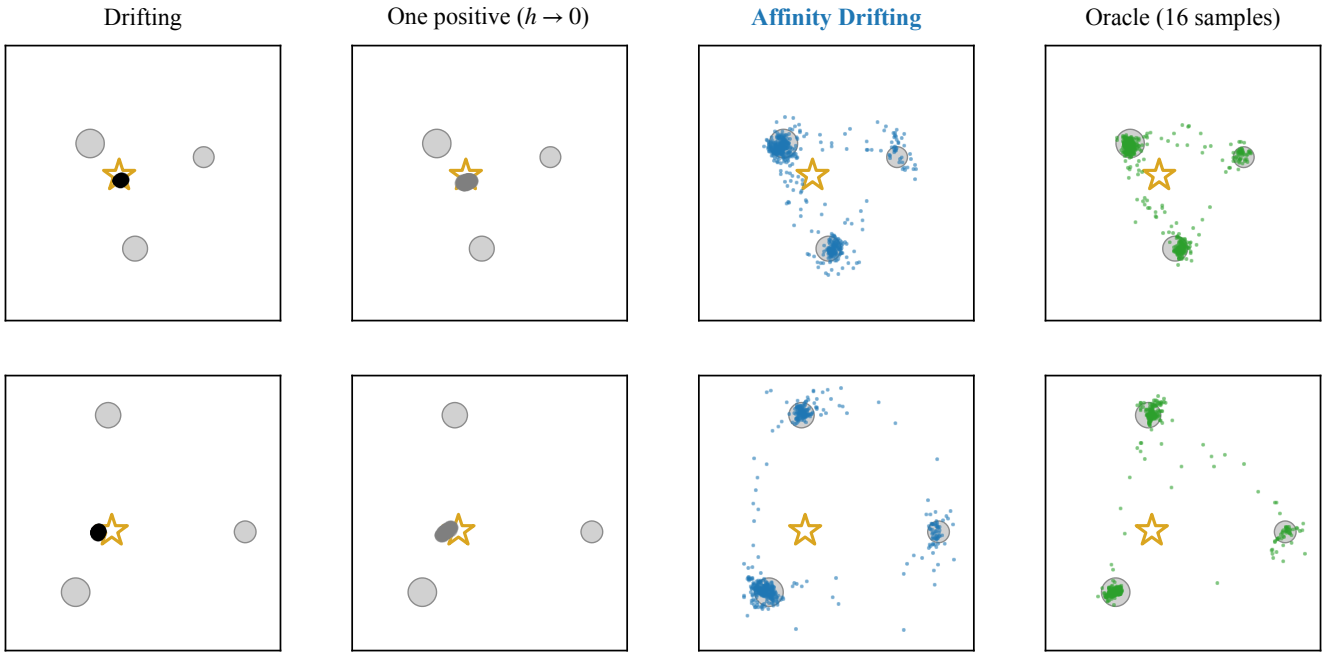


Figure 1. **Mean collapse with one sample per condition.** Samples (512 per panel) from generators trained on a 2-D toy in which each training condition c has exactly one sample, shown for two held-out conditions. Grey: the true $p_{\text{data}}(\cdot \mid c)$, a three-component Gaussian mixture; hollow $\star$: the conditional mean $\mathbb{E}[\mathbf{x} \mid c]$. The original Drifting objective and the one-positive objective place every sample at the conditional mean, between the modes; an oracle given many true samples per condition covers the modes. Affinity Drifting uses only the single training samples, pooled across similar conditions, and recovers the modes. Seed 0 of five.

ished experiments and marks the others as pending.

2. Method

Drifting. A generator f_θ maps noise $\boldsymbol{\epsilon} \sim p_{\boldsymbol{\epsilon}}$ and a condition $c \in \mathcal{C}$ to a sample $\mathbf{x} = f_\theta(\boldsymbol{\epsilon}, c) \in \mathbb{R}^D$; its conditional pushforward is $q_\theta(\cdot \mid c) = f_\theta(\cdot, c)_{\#} p_{\boldsymbol{\epsilon}}$. Following [2], the drifting field is $\mathbf{V}_{p,q}(\mathbf{x}) = \mathbf{V}_p^+(\mathbf{x}) - \mathbf{V}_q^-(\mathbf{x})$ with

$$\mathbf{V}_p^+(\mathbf{x}) = \frac{1}{Z_p(\mathbf{x})} \mathbb{E}_p\big[k(\mathbf{x}, \mathbf{y}^+)(\mathbf{y}^+ - \mathbf{x})\big], \tag{1}$$

$Z_p(\mathbf{x}) = \mathbb{E}_p[k(\mathbf{x}, \mathbf{y}^+)]$, and $\mathbf{V}_q^-$, Z_q the same expressions over $\mathbf{y}^- \sim q$. With $p = p_{\text{data}}(\cdot \mid c)$ and $q = q_\theta(\cdot \mid c)$ the generator is trained with

$$\mathcal{L} = \mathbb{E}_{c,\boldsymbol{\epsilon}}\Big[\big\| f_\theta(\boldsymbol{\epsilon}, c) - \texttt{stopgrad}\big(f_\theta(\boldsymbol{\epsilon}, c) + \mathbf{V}_{p,q_\theta}(f_\theta(\boldsymbol{\epsilon}, c))\big)\big\|^2\Big]. \tag{2}$$

In practice $\mathbf{V}$ is replaced by the normalized drift $\tilde{\mathbf{V}} = \mathbf{V}/\lambda$ with $\lambda = \sqrt{\mathbb{E}\big[\frac{1}{D}\|\mathbf{V}\|^2\big]}$ over the batch, summed over several kernel scales [2].

Let the training set be $\{(c_i, \mathbf{y}_i)\}_{i=1}^N$ with distinct conditions, so that the empirical conditional at c_i is the Dirac measure $\delta_{\mathbf{y}_i}$.

Lemma 1 (One positive is regression)**.** *If $p = \delta_{\mathbf{y}}$, then $\mathbf{V}_p^+(\mathbf{x}) = \mathbf{y} - \mathbf{x}$ for every kernel $k > 0$. Hence, if $p_{\text{data}}(\cdot \mid c) = \delta_{\mathbf{y}_c}$ for every c, (2) without negatives equals $\mathbb{E}_{c,\boldsymbol{\epsilon}}\|f_\theta(\boldsymbol{\epsilon}, c) - \mathbf{y}_c\|^2$, and so does its gradient.*

Proofs are in Appendix A.

Affinity Drifting. We draw positives from the Nadaraya–Watson estimate of the conditional distribution [8, 10],

$$\hat{p}_h(\cdot \mid c) = \sum_{i=1}^N \tilde{w}_i(c; h)\, \delta_{\mathbf{y}_i}, \qquad \tilde{w}_i(c; h) = \frac{w_i(c; h)}{\sum_{j=1}^N w_j(c; h)}, \tag{3}$$

$$w_i(c; h) = \exp\Big(-\frac{\|\psi(c) - \psi(c_i)\|^2}{2h^2}\Big),$$

with a fixed condition embedding ψ (the true coordinates in the toy; a frozen text encoder in later stages), and keep (1) with $p = \hat{p}_h(\cdot \mid c)$. Negatives are drawn from the analogous estimate $\hat{q}_{h'}(\cdot \mid c) = \sum_i \tilde{w}_i(c; h')\, q_\theta(\cdot \mid c_i)$ for $h' > 0$, and from $\hat{q}_0(\cdot \mid c) = q_\theta(\cdot \mid c)$ by default. The bandwidth $h = h(c)$ is set per condition to the smallest value whose effective sample size [4] $n_{\text{eff}}(c) = \big(\sum_i w_i(c; h)\big)^2 / \sum_i w_i(c; h)^2$ reaches $n^\star = 16$, so every condition pools equally many positives. The limit $h \to 0$ recovers the one-positive objective, and $h, h' \to \infty$ unconditional drifting.

Fixed points. For the Gaussian kernel

$$k(\mathbf{x}, \mathbf{y}) = \exp\Big(- \frac{\|\mathbf{x} - \mathbf{y}\|^2}{2\sigma^2} \Big)$$

the field is a difference of smoothed scores.

Lemma 2 (Score difference)**.** *For the Gaussian kernel,*

$$\mathbf{V}_{p,q}(\mathbf{x}) = \sigma^2 \, \nabla_{\mathbf{x}} \big[\log Z_p(\mathbf{x}) - \log Z_q(\mathbf{x}) \big],$$

and $\mathbf{V}_{p,q} \equiv \mathbf{0}$ *if and only if* $q = p$.

Proposition 1 (Equilibria)**.** *For the Gaussian kernel, (i) with* $h' = 0$, $\mathbf{V} \equiv \mathbf{0}$ *at* c *if and only if* $q_\theta(\cdot \mid c) = \hat{p}_h(\cdot \mid c)$*; (ii) with* $h' = h$*, memorization of the training positives,* $q_\theta(\cdot \mid c_i) = \delta_{\mathbf{y}_i}$ *for all* i*, gives* $\mathbf{V} \equiv \mathbf{0}$ *at every* c*, whereas, for* $N \geq 2$*,* $h < \infty$ *and distinct* $\mathbf{y}_i$ *and* $\psi(c_i)$*, the Nadaraya–Watson mixtures* $q_\theta(\cdot \mid c_i) = \hat{p}_h(\cdot \mid c_i)$ *do not.*

The negative bandwidth h' thus selects the equilibrium. We use three Gaussian scales and compute the drift normalization for the whole field, as in [2]; each scale satisfies Lemma 2, so the equilibria are unchanged (Appendix B).

3. Experiments

Protocol. Each comparison changes a single factor of a fixed base configuration; a run does not start if its configuration differs from the declared change. Predictions, falsification criteria and decision rules are committed before any run, and every outcome (pass, fail or inconclusive) is computed from those rules only. All methods share the seeds (five per experiment), the evaluation conditions and the learning rate, which is selected once on an unconditional model. Intervals are 95% paired bootstrap confidence intervals over seeds (10^4 resamples). Every experiment also trains the original Drifting objective, unchanged, as a reference that is reported but not judged.

3.1. Fixed points in a particle system

Without a network, each of 32 conditions on a line holds 256 particles updated by $\mathbf{x} \leftarrow \mathbf{x} + \eta \, \mathbf{V}(\mathbf{x})$, a generator of unlimited capacity, with one positive per condition. Details are in Appendix B. The field agrees with the automatic-differentiation score difference of Lemma 2 to 1.6×10^{-15} in float64 (PASS). Increasing h' from 0 to h moves the equilibrium monotonically from the Nadaraya–Watson mixture to memorization (Fig. 2; PASS), whereas the original Drifting objective memorizes outright (share of particles nearest their own positive: 0.951).

The assignment of particles to individual positives, however, misses the pre-registered tolerance for the mixture (total variation 0.147 [0.145, 0.149] against 0.1; FAIL): with a kernel wider than the spacing of the positives, the field cannot resolve which nearby positive a particle sits on. Compared at the resolution of the kernel — total variation between the σ-smoothed densities, an analysis that was not pre-registered — the equilibrium matches $\hat{p}_h$ to 1.0×10^{-4}. We also draw negatives from another condition's data with weight γ, a two-condition analogue of the mixing rate γ of guidance in [2]. For Gaussian conditionals the predicted equilibrium is Gaussian in closed form, and the particle system reaches it under a Gaussian moment closure (relative error 3.3×10^{-6}). With finite samples, however, it is stable only for small weights (error 0.0278 at $\gamma = 0.25$, divergence from $\gamma = 0.5$; FAIL).

Method	$W_2 \downarrow$	$\operatorname{tr} \boldsymbol{\Sigma}_q / \operatorname{tr} \boldsymbol{\Sigma}_p$	Prec. ↑	Rec. ↑	Modes ↑
Drifting	1.4	2.8×10^{-5}	0.0023	0	0.34
One positive ($h \to 0$)	1.4	3.8×10^{-4}	0.0017	1.5×10^{-6}	0.35
Constant sample kernel	1.3	0.012	0.0022	0.007	0.48
Random neighbors	1.1	1.3	0.1	0.97	1
Noisy positive	1.4	4.4×10^{-4}	1.2×10^{-5}	0	0.36
Affinity Drifting	0.47	0.97	0.74	0.94	1
Oracle (16 samples)	0.5	0.91	0.83	0.78	0.98

Table 1. **2-D toy, held-out conditions** (mean over five seeds). Above the line: the one-positive objectives and three ablations of Affinity Drifting; below: Affinity Drifting and an oracle that sees 16 true samples per condition and step.

3.2. Mean collapse and its repair in a 2-D toy

Setup. Each of $N = 10{,}000$ training conditions $c \in [0, 1]^2$ has one sample from $p_{\text{data}}(\cdot \mid c)$, a three-component Gaussian mixture whose components rotate with c_1 and whose radius and weights change with c_2, so the conditional mean lies between modes. An MLP generator $f_\theta(\boldsymbol{\epsilon}, c)$ sees c quantized to cells of width $\Delta = 1/16$, about 39 training conditions per cell, and is trained with Adam for 20,000 steps at a learning rate selected once on the unconditional model. We evaluate 512 samples on each of 256 held-out conditions and report the Wasserstein-2 distance W_2 to the true conditional, the ratio of total variances $\operatorname{tr} \boldsymbol{\Sigma}_q / \operatorname{tr} \boldsymbol{\Sigma}_p$ (1 for a correct spread, 0 for a point mass), precision and recall [5], the fraction of modes captured, and the fraction of off-support samples. Data, architecture, training and metrics are specified in Appendix B.

One positive collapses. Both the original Drifting objective and the one-positive objective ($h \to 0$) collapse every condition to a point at the conditional mean (variance ratio 2.8×10^{-5} $[2.0 \times 10^{-5}, 3.4 \times 10^{-5}]$ and 3.8×10^{-4} $[2.9 \times 10^{-4}, 4.7 \times 10^{-4}]$; off-support share 1), while the oracle, given 16 true samples per condition and step, covers the modes (variance ratio 0.909 [0.906, 0.915], modes captured 0.977; PASS).

Pooling positives restores the mixture. Trained on the same single positives, Affinity Drifting captures every mode

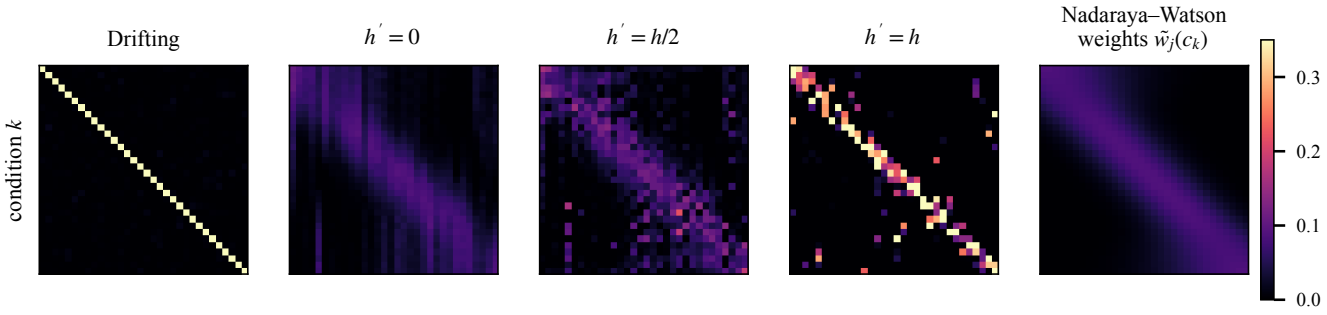


Figure 2. **The negative bandwidth h' selects the equilibrium** (particle system, mean over seeds). Entry (k, j): share of condition k's particles nearest to positive $\mathbf{y}_j$. The original Drifting objective memorizes (diagonal). With $h' = 0$, mass spreads over the pooled neighbors as the Nadaraya–Watson weights $\tilde{w}_j(c_k)$ predict (right); larger h' narrows the band toward memorization.

(fraction 1, recall 0.94) and reaches a held-out W_2 of 0.469 $[0.46, 0.477]$ against 1.41 $[1.41, 1.42]$ for the one-positive objective (ratio 0.33; Tab. 1), on par with the oracle. The preregistered comparison is PASS: Affinity Drifting improves on the one-positive objective and on every ablation, with paired confidence intervals that exclude zero. The ablations fail in the ways the mechanism predicts. With a constant sample-space kernel, the same pooled positives give an $\mathbf{x}$-independent target, their weighted average, and the generator collapses as with one positive (variance ratio 0.0122). Pooling as many random neighbors spreads samples over all modes (fraction 1) but places most of them off the conditional support (off-support share 0.922). Perturbing the single positive with noise of matched variance still collapses (variance ratio 4.4×10^{-4}). The gain therefore comes from the $\mathbf{x}$-dependent assignment of samples to the modes of semantically similar positives, not from more or smoother targets. Figure 3 shows this assignment: with one positive every attraction vector points at the same target, whereas the pooled positives partition the plane into one basin per mode. We see no sign of memorization: the share of samples within half the median nearest-neighbor spacing of some training positive is 0.165 on training conditions, the same as for true samples of $p_{\text{data}}(\cdot \mid c)$ (0.166, an analysis that was not preregistered), whereas the one-positive objective, whose samples sit between the modes, scores 0.00443.

Bandwidth. Scaling the bandwidth $h(c)$ selected by $n_{\text{eff}} = n^\star$ by a factor $m \in \{\frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8\}$ gives a U-shaped held-out W_2 with its minimum at $m = 1$ (PASS): 0.925 at $m = \frac{1}{4}$, 0.469 at $m = 1$ and 0.77 at $m = 8$. Too small a bandwidth pools too few positives to resolve the modes; too large a bandwidth pools conditions whose modes lie elsewhere. A single bandwidth shared by all conditions, set to the median of the selected $h(c)$, performs alike in this toy (0.467), where conditions are uniformly dense. The unnormalized drift energy $\mathbb{E}\|\mathbf{V}_{p,q}\|^2$ after training decreases monotonically in h (0.25 at $m = 1$, 0.125 at $m = 8$): a larger bandwidth gives a smoother target that is easier to match, not a better model. The drift energy is therefore not a criterion for selecting h; we never use it for selection.

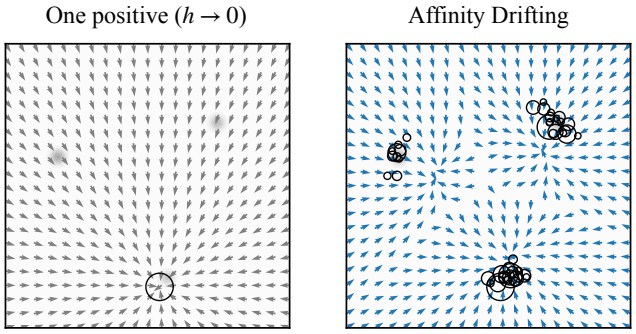


Figure 3. **Attraction field.** Direction of $\mathbf{V}_p^+(\mathbf{x})$ for one training condition c. One positive ($p = \delta_{\mathbf{y}}$): every vector points at the single target. Affinity Drifting ($p = \hat{p}_h(\cdot \mid c)$): vectors point to the nearest mode of the pooled positives (circles, area $\propto \tilde{w}_i(c)$).

Conditional resolution does not explain the gap. We predicted that the gap between the one-positive objective and Affinity Drifting grows as the generator's resolution in c is coarsened by quantization, and vanishes on training conditions without quantization, where a generator could in principle fit each positive. The prediction fails (FAIL; Fig. 4). On training conditions the W_2 of the one-positive objective is 1.41 without quantization and 1.41 at $\Delta = 1/16$, against 0.448 and 0.479 for Affinity Drifting: the gap does not shrink at $\Delta = 0$, and it narrows at the coarsest $\Delta = 1/4$ (1.43 against 0.741) because Affinity Drifting degrades there, not because the one-positive objective improves. Even without quantization, the one-positive generator does not fit its training positives: its variance ratio on training conditions is 4.5×10^{-4}, and in an analysis that was not preregistered, the mean of its samples lies 0.098 from the conditional mean $\mathbb{E}[\mathbf{x} \mid c]$ and 1.51 from the condition's own positive $\mathbf{y}_i$. The network's own smoothness in c, not the

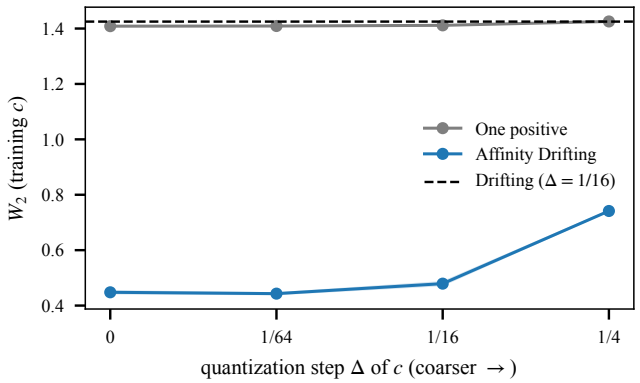


Figure 4. **Conditional resolution.** W_2 on training conditions of the one-positive objective and of Affinity Drifting as the quantization of c seen by the generator coarsens (mean over seeds, 95% confidence interval); dashed: the original Drifting objective at $\Delta = 1/16$. The predicted convergence of the two at $\Delta = 0$ does not occur.

quantization step, thus sets its effective resolution in this range: an MLP trained for 20,000 steps on 10,000 conditions regresses to the conditional mean whether or not c is quantized. Collapse is therefore the default behavior of the one-positive objective here, and explicit quantization from $\Delta = 0$ to $1/16$ does not change it.

A smooth version of the same test, with spectral normalization constraining the condition embedding g to be L-Lipschitz for $L \in \{64, 16, 4, 1\}$, gives the same picture and is INCONCLUSIVE under its pre-registered rule: the W_2 of the one-positive objective is 1.44 at $L = 64$ and 1.42 at $L = 1$, against 0.445 and 0.418 for Affinity Drifting. The constraint, however, does not bind. It acts on g only, and the unconstrained layers that follow can rescale $g(c)$, so f_θ itself is not L-Lipschitz in c. Even at $L = 1$, Affinity Drifting is no worse than at $L = 64$, although the mode centers of $p_{\text{data}}(\cdot \mid c)$ move at rate $2\pi(1 + c_2)$ in c_1. This experiment therefore does not test the dependence on capacity.

The negative bandwidth does not select the equilibrium of a trained generator. In the particle system, h' moved the equilibrium from the Nadaraya–Watson mixture ($h' = 0$) to memorization ($h' = h$). We predicted the same for the trained generator: more memorization of training positives with $h' = h$, and, with $h' = 0$, a generated distribution within total variation 0.1 of the Nadaraya–Watson mixture, both smoothed at the finest kernel scale. The prediction fails (FAIL). The two settings are nearly indistinguishable: memorization share 0.166 with $h' = 0$ and 0.168 with $h' = h$ (the difference is positive but small, and both equal the share for true samples, 0.166), share of samples at the condition's own positive 4.8×10^{-4} and 5.2×10^{-4}, and W_2 0.458 and 0.466 on training conditions. The total variation to the mixture is 0.386 with $h' = 0$. In an analysis that was not pre-registered, the true conditional $p_{\text{data}}(\cdot \mid c)$ is at 0.359 from the same mixture (finite-sample floor 0.0893): the generator is about as far from the mixture of the 16 pooled points as the true conditional is, and close to the latter in W_2. As with quantization, the trained generator does not reach the non-parametric fixed points; it interpolates between nearby conditions and approximates $p_{\text{data}}(\cdot \mid c)$ rather than the discrete mixture or the training positives.

4. Status, deviations and limitations

This is a working draft; Tab. 2 lists every pre-registered experiment and its outcome at the time of the build. Changes to the plan are logged with reasons and approvals before any affected run; those that matter for reading the results are the following. The learning-rate selection failed twice before passing: first because the unconditional model was misspecified (negatives from each condition's samples rather than from the whole batch), then narrowly. A single kernel scale transported mass too locally and was replaced by three scales. 64 instead of 8 samples per condition are generated, since negatives estimated from 8 samples are strongly biased. The drift normalization was changed from separate normalization of $\mathbf{V}_p^+$ and $\mathbf{V}_q^-$ to normalization of the whole field, as in [2], after the separate version sent one-positive samples outside the data. Runs made before a correction are kept and excluded from all results.

The toy uses the true condition coordinates as the embedding ψ; text embeddings, learned features and real captions are later stages of the study and are not reported here.

Setting	Pre-registered claim	Outcome
Particles	Mean shift equals the scaled score difference	PASS
Particles	Negatives with $h' = h$ memorize the positives	INCONCLUSIVE
Particles	Negatives with $h' = 0$: Nadaraya–Watson equilibrium	FAIL
Particles	Equilibrium moves monotonically with h'	PASS
Particles	Cross-condition negatives: closed-form equilibrium	FAIL
Particles	Same, under a Gaussian moment closure	PASS
Particles	Joint vs. separate softmax: same equilibrium	FAIL
2-D toy	Unconditional model fits the marginal (learning rate)	FAIL / FAIL / PASS
2-D toy	One positive, no negatives: ℓ_2 regression	PASS
2-D toy	One positive regresses to the conditional mean	PASS
2-D toy	Affinity Drifting restores the mixture; beats ablations	PASS
2-D toy	Gap grows as resolution in c coarsens (quantization)	FAIL
2-D toy	Same, with a Lipschitz constraint on c	INCONCLUSIVE
2-D toy	h' selects the equilibrium of a trained generator	FAIL
2-D toy	Bandwidth: U-shaped error, minimum near $n^\star$	PASS

Table 2. Pre-registered experiments and their outcomes at build time.

5. Related work

Affinity Drifting builds on drifting models [2], which turn kernel mean-shift [1] into a training signal for one-step generators. Its positive distribution is a Nadaraya–Watson kernel estimate [8, 10] of the conditional distribution, with the

bandwidth set by the effective sample size of the weights [4]. The field is closely related to gradients of kernel two-sample statistics (MMD) [3], which generative models have combined with mixtures of kernel scales for the same reason we do [6]. The evaluation of mode collapse follows standard practice: precision and recall [5] and density and coverage [9]. Capacity is controlled by quantizing the condition and by spectral normalization [7].

References

[1] Dorin Comaniciu and Peter Meer. Mean shift: A robust approach toward feature space analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(5):603–619, 2002. 5

[2] Deng, Li, Li, Du, and He. Generative modeling via drifting. *arXiv preprint arXiv:2602.04770*, 2026. 1, 2, 3, 5, 6

[3] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012. 6

[4] Leslie Kish. *Survey Sampling*. Wiley, 1965. 2, 6

[5] Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. In *Advances in Neural Information Processing Systems*, 2019. 3, 6, 7

[6] Chun-Liang Li, Wei-Cheng Chang, Yu Cheng, Yiming Yang, and Barnabás Póczos. MMD GAN: Towards deeper understanding of moment matching network. In *Advances in Neural Information Processing Systems*, 2017. 6

[7] Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral normalization for generative adversarial networks. In *International Conference on Learning Representations*, 2018. 6

[8] E. A. Nadaraya. On estimating regression. *Theory of Probability and Its Applications*, 9(1):141–142, 1964. 2, 5

[9] Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *International Conference on Machine Learning*, 2020. 6

[10] Geoffrey S. Watson. Smooth regression analysis. *Sankhyā: The Indian Journal of Statistics, Series A*, 26(4):359–372, 1964. 2, 5

A. Proofs

Throughout, p and q are Borel probability measures on $\mathbb{R}^D$ and $k > 0$, so that $Z_p, Z_q > 0$ and the fields are well defined.

Proof of Lemma 1. For $p = \delta_{\mathbf{y}}$, $Z_p(\mathbf{x}) = k(\mathbf{x}, \mathbf{y}) > 0$ and $\mathbb{E}_p[k(\mathbf{x}, \mathbf{y}^+)(\mathbf{y}^+ - \mathbf{x})] = k(\mathbf{x}, \mathbf{y})(\mathbf{y} - \mathbf{x})$, so $\mathbf{V}_p^+(\mathbf{x}) = \mathbf{y} - \mathbf{x}$. Without negatives $\mathbf{V}_{p,q_\theta} = \mathbf{V}_p^+$, hence at $\mathbf{x} = f_\theta(\boldsymbol{\epsilon}, c)$ the argument of `stopgrad` in (2) has value $f_\theta(\boldsymbol{\epsilon}, c) + \mathbf{y}_c - f_\theta(\boldsymbol{\epsilon}, c) = \mathbf{y}_c$, and the integrand equals $\|f_\theta(\boldsymbol{\epsilon}, c) - \mathbf{y}_c\|^2$. Since `stopgrad` blocks the gradient through its argument, $\nabla_\theta \|f_\theta - \texttt{stopgrad}(\boldsymbol{u})\|^2 = 2\,(\partial_\theta f_\theta)^\top (f_\theta - \boldsymbol{u})$ evaluated at $\boldsymbol{u} = \mathbf{y}_c$, which is $\nabla_\theta \|f_\theta(\boldsymbol{\epsilon}, c) - \mathbf{y}_c\|^2$. □

Proof of Lemma 2. Let φ_σ be the density of $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$. Then $k(\mathbf{x}, \mathbf{y}) = (2\pi\sigma^2)^{D/2} \varphi_\sigma(\mathbf{x} - \mathbf{y})$, so $Z_p(\mathbf{x}) = (2\pi\sigma^2)^{D/2} (p * \varphi_\sigma)(\mathbf{x})$. Moreover $\nabla_{\mathbf{x}} k(\mathbf{x}, \mathbf{y}) = \sigma^{-2} k(\mathbf{x}, \mathbf{y})(\mathbf{y} - \mathbf{x})$ and $\|k(\mathbf{x}, \mathbf{y})(\mathbf{y} - \mathbf{x})\| \le \sup_{r \ge 0} r e^{-r^2/(2\sigma^2)} = \sigma e^{-1/2}$, so differentiation under the expectation is justified by dominated convergence: $\nabla_{\mathbf{x}} Z_p(\mathbf{x}) = \sigma^{-2}\, \mathbb{E}_p[k(\mathbf{x}, \mathbf{y}^+)(\mathbf{y}^+ - \mathbf{x})]$. Dividing by $Z_p(\mathbf{x})$ gives $\mathbf{V}_p^+ = \sigma^2 \nabla_{\mathbf{x}} \log Z_p$, likewise $\mathbf{V}_q^- = \sigma^2 \nabla_{\mathbf{x}} \log Z_q$, and the identity follows.

If $q = p$ the field vanishes trivially. Conversely, $\mathbf{V}_{p,q} \equiv \mathbf{0}$ means $\nabla_{\mathbf{x}} \log(Z_p / Z_q) \equiv \mathbf{0}$ on the connected set $\mathbb{R}^D$, so $Z_p = \kappa Z_q$ for a constant $\kappa > 0$. Integrating over $\mathbb{R}^D$, $\int Z_p \, d\mathbf{x} = (2\pi\sigma^2)^{D/2} = \int Z_q \, d\mathbf{x}$, so $\kappa = 1$ and $p * \varphi_\sigma = q * \varphi_\sigma$. Taking characteristic functions, $\hat{p}(\boldsymbol{\omega}) e^{-\sigma^2 \|\boldsymbol{\omega}\|^2/2} = \hat{q}(\boldsymbol{\omega}) e^{-\sigma^2 \|\boldsymbol{\omega}\|^2/2}$ for all $\boldsymbol{\omega}$, hence $\hat{p} = \hat{q}$ and $p = q$ by the uniqueness theorem for characteristic functions. □

Proof of Proposition 1. (i) With $h' = 0$ the positives at c follow $\hat{p}_h(\cdot \mid c)$ and the negatives $q_\theta(\cdot \mid c)$; Lemma 2 gives $\mathbf{V} \equiv \mathbf{0}$ if and only if $q_\theta(\cdot \mid c) = \hat{p}_h(\cdot \mid c)$.

(ii) With $h' = h$ and $q_\theta(\cdot \mid c_i) = \delta_{\mathbf{y}_i}$, the negatives at any c follow $\hat{q}_h(\cdot \mid c) = \sum_i \tilde{w}_i(c; h)\, \delta_{\mathbf{y}_i} = \hat{p}_h(\cdot \mid c)$, so positives and negatives coincide and $\mathbf{V} \equiv \mathbf{0}$ by Lemma 2. Now let $q_\theta(\cdot \mid c_i) = \hat{p}_h(\cdot \mid c_i)$, assume the $\mathbf{y}_i$ are distinct, the $\psi(c_i)$ are distinct, $N \ge 2$ and $h < \infty$, and write $W_{kj} = \tilde{w}_j(c_k; h)$. At $c = c_k$ the negatives follow $\sum_i W_{ki} \hat{p}_h(\cdot \mid c_i) = \sum_j (W^2)_{kj} \delta_{\mathbf{y}_j}$ and the positives $\sum_j W_{kj} \delta_{\mathbf{y}_j}$; since the $\mathbf{y}_j$ are distinct, $\mathbf{V} \equiv \mathbf{0}$ at every c_k requires $W^2 = W$. W is row-stochastic with positive entries, so by the Perron–Frobenius theorem the eigenvalue 1 is simple and every other eigenvalue has modulus below 1; an idempotent matrix has eigenvalues in $\{0, 1\}$, so W would have rank one, *i.e.* identical rows. Rows $k \ne l$ are identical only if W_{kj}/W_{lj} does not depend on j, *i.e.* only if $\langle \psi(c_k) - \psi(c_l), \psi(c_j) \rangle$ does not depend on j. This fails for $j = k$ against $j = l$, where the two values differ by $\|\psi(c_k) - \psi(c_l)\|^2 > 0$. Hence the Nadaraya–Watson mixtures are not a fixed point. □

B. Implementation and experimental details

Kernels and normalization. The sample-space kernel is Gaussian at three scales $\sigma \in \{0.071, 0.384, 2.07\}$, spaced geometrically from the median $n^\star$-th-nearest-neighbor distance among the positives to the median pairwise distance (the median heuristic [3]), and fixed once per experiment. For each scale the field $\mathbf{V}_{p,q}$ is normalized as a whole, $\tilde{\mathbf{V}}_\sigma = \mathbf{V}_\sigma / \lambda_\sigma$ with $\lambda_\sigma = \sqrt{\mathbb{E}[\frac{1}{D}\|\mathbf{V}_\sigma\|^2]}$ over the batch, and the normalized fields are summed, as in the multi-temperature drift of [2]. Normalizing $\mathbf{V}_p^+$ and $\mathbf{V}_q^-$ separately, which we used first, keeps both terms at unit scale even where they should cancel, and sent a few percent of one-positive samples far outside the data (Sec. 4). A sample is excluded

from its own negatives. The per-condition bandwidth $h(c)$ is found by bisection in $\log h$ (relative tolerance 10^{-10}). The baseline uses the original Drifting objective unchanged: kernel $k(\mathbf{x}, \mathbf{y}) = \exp(-\|\mathbf{x}-\mathbf{y}\|/\tau)$ with $\tau \in \{0.02, 0.05, 0.2\}$, its softmax normalization over positives and negatives, distances scaled by batch statistics, and one normalization per temperature. We use the Gaussian kernel for our method so that Lemma 2 applies.

Particle system. $K = 32$ conditions with $\psi(c_k)$ evenly spaced on $[0, 1]$ each have one positive $\mathbf{y}_k \sim \mathcal{U}([-1, 1]^2)$, drawn once with minimum separation 0.15 and shared by all seeds. Each condition holds 256 particles initialized from $\mathcal{U}([-1.5, 1.5]^2)$ and updated by $\mathbf{x} \leftarrow \mathbf{x} + \eta\,\mathbf{V}(\mathbf{x})$ with $\eta = 0.5$ and the unnormalized field; the negatives of a particle are all particles of its condition. The kernel has a single scale, half the median pairwise distance of the positives. Integration stops when the root-mean-square field norm stays below 10^{-6} for 1,000 consecutive steps, or after 50,000 steps. Lemma 2 is checked against automatic differentiation in float64. For the guidance analogue, $p_{\text{data}}(\cdot \mid c) = \mathcal{N}(\boldsymbol{\mu}_1, s^2\mathbf{I})$ and $p_{\text{data}}(\cdot \mid c') = \mathcal{N}(\boldsymbol{\mu}_2, s^2\mathbf{I})$ with $\boldsymbol{\mu}_1 = (1, 0)$, $\boldsymbol{\mu}_2 = (-1, 0)$, $s = 0.5$ and 2,048 positives each; the negatives of c add the data of c' with weight γ.

2-D toy data. Conditions are $c \sim \mathcal{U}([0, 1]^2)$, with $N = 10{,}000$ training conditions (one positive each), 256 held-out conditions, and 256 of the training conditions used for evaluation. $p_{\text{data}}(\cdot \mid c)$ is a mixture of three Gaussians $\mathcal{N}(\boldsymbol{\mu}_j(c), 0.1^2\mathbf{I})$, $j \in \{0, 1, 2\}$, with centers at angle $2\pi c_1 + 2\pi j/3$ and radius $1 + c_2$, and weights $\pi_j(c) \propto \exp\big(\beta(c_2 - \frac{1}{2})(j-1)\big)$ with $\beta = 2$.

Generator and training. $f_\theta(\boldsymbol{\epsilon}, c)$ is an MLP on the concatenation of $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_4)$ and a condition embedding $g(c) \in \mathbb{R}^{64}$ (two linear layers with a ReLU), with three hidden layers of width 256, SiLU activations, and a linear output in $\mathbb{R}^2$. Quantization replaces c by the center $(\lfloor c/\Delta \rfloor + \frac{1}{2})\Delta$ of its cell, componentwise; $\Delta = 1/16$ unless stated otherwise. The Lipschitz variant applies spectral normalization to both layers of g and multiplies its output by L. We train with Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, no weight decay) at a constant learning rate of 10^{-3} for 20,000 steps, with 8 conditions $\times$ 64 generated samples per batch, and evaluate an exponential moving average of the weights (decay 0.999). The learning rate was selected once among $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}$ on the unconditional model, whose marginal Wasserstein-2 distance is 0.136 $[0.129, 0.144]$ against a finite-sample floor of 0.0876, and shared by every method. The negative-bandwidth experiment uses $\Delta = 0$, three times as many steps, and batches of one condition with seven neighbors drawn with probabilities $\tilde{w}_i(c; h)$, so that pooled negatives are available.

Metrics. Each condition is evaluated with 512 generated and 512 true samples. W_2 is computed exactly by optimal assignment. The variance ratio is $\operatorname{tr}\boldsymbol{\Sigma}_q / \operatorname{tr}\boldsymbol{\Sigma}_p$ of the sample covariances. Precision and recall [5] use $k = 3$ nearest neighbors. A sample belongs to the mode with the nearest center, and a mode counts as captured if it receives at least 5% of the samples. A sample is off-support if its true log-density is below the 1% quantile of the log-densities of the true samples of that condition. A sample is memorized if it lies within ε of a training positive, with ε half the median nearest-neighbor distance among the training positives. The total variation to the Nadaraya–Watson mixture compares both distributions smoothed at the finest kernel scale. Results are means over five seeds with 95% bootstrap confidence intervals (10^4 resamples), paired across methods.